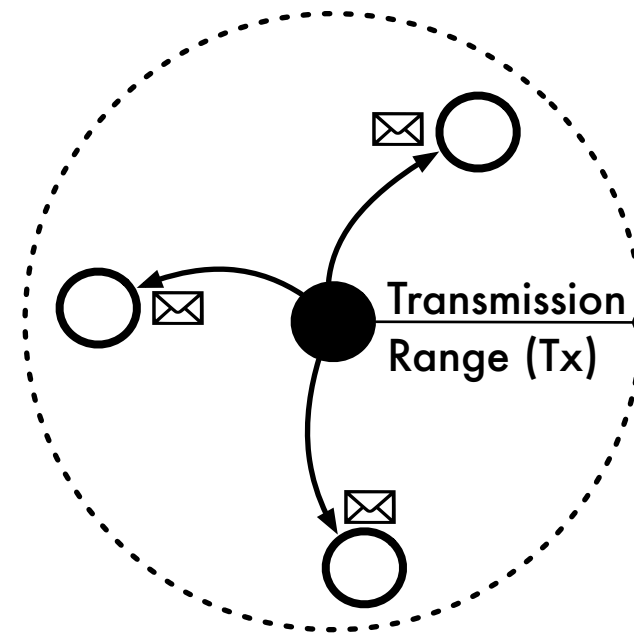


Mobile Ad hoc Networks (MANETs)

MANETs [1] are autonomous systems that do not rely on fixed infrastructures. Nodes move freely and forward packets in a peer-to-peer way. Suitable scenarios are:

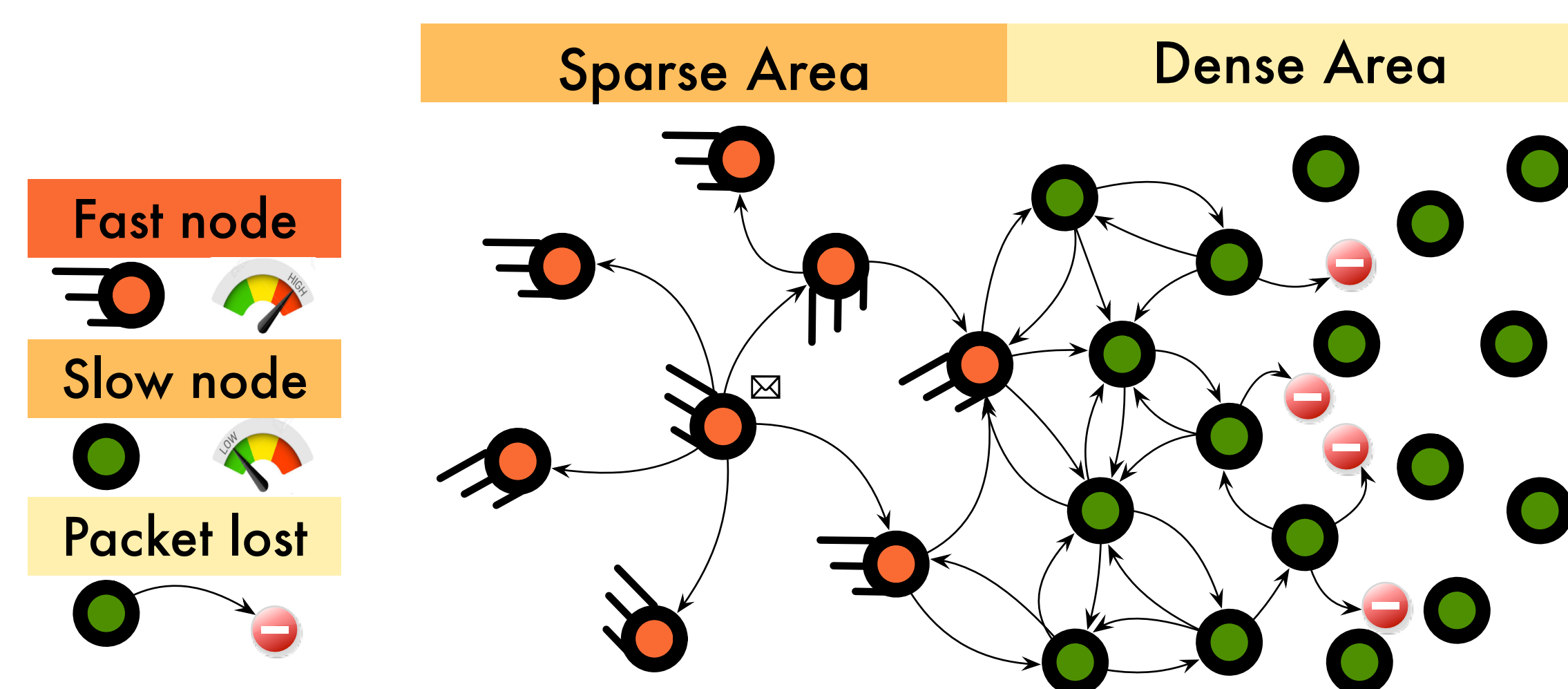
- vehicular area networks
- monitoring of zones in danger
- networks in rural regions



Broadcast is a key operation in MANETs but far of being easy. Find a **forwarding set**, in a decentralised manner, such that:

- nodes in this set retransmit incoming packets
- other nodes act only as receivers
- nodes deal with mobility and density while:
 - increasing network coverage
 - avoiding to drain nodes batteries

Broadcast protocols are overlay-based or controlled flooding [2]. In a system where nodes move freely, one approach may result in lost of packets:



Likelihood of collisions and contentions increase in dense area. Our study [3] reveals that fits better:

- an overlay-based protocol in dense/static areas
- a controlled flooding protocol where nodes move faster

Research Questions

- How nodes adapt their broadcast protocol?
- Which information is useful to foster a will of change?

Adaptive approach to broadcast in MANETs

Problem. Broadcast in an heterogeneous system:

- where mobile devices are carried by humans
- where density and velocity of nodes varies
- maintaining high reachability
- reducing overall energy consumption

Objective. Design and evaluation of adaptive-based approaches to broadcast in MANETs

Challenges in adaptation at run-time:

- specification of adaptation rules
- decision of change: collective or individual?
- system monitoring
 - remaining energy, neighbours number, signal strength, ...
- avoid system degradation

Our hybrid solution to broadcast

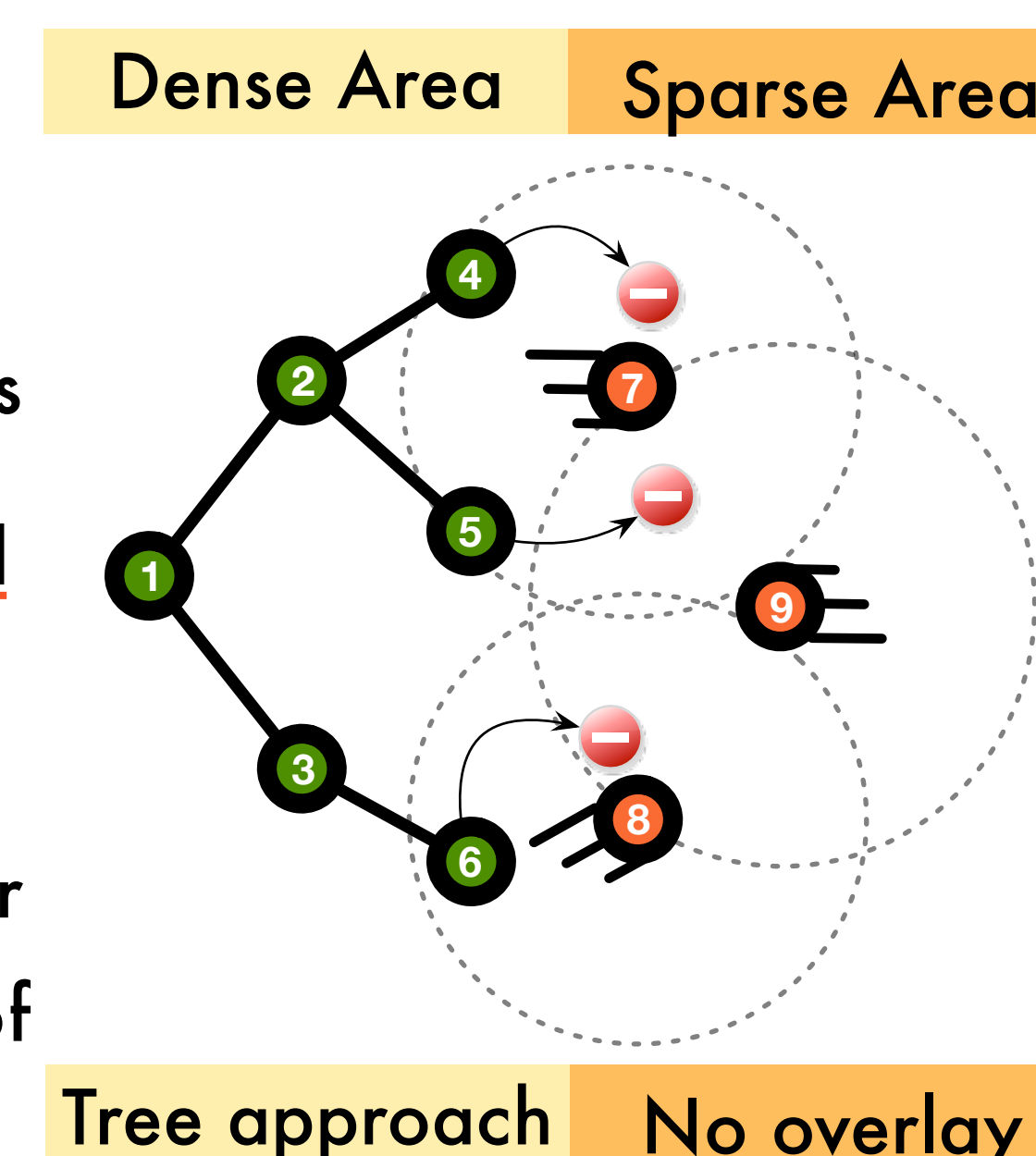
Overlay-based in dense/static Controlled flooding in sparse/dynamic region

Interoperability issues with 2 running protocols

- Source node: 1
 - coverage 66%
- Role of nodes 4, 5 & 6 matters
 - leafs nodes do not forward

Our Interoperability mechanism:

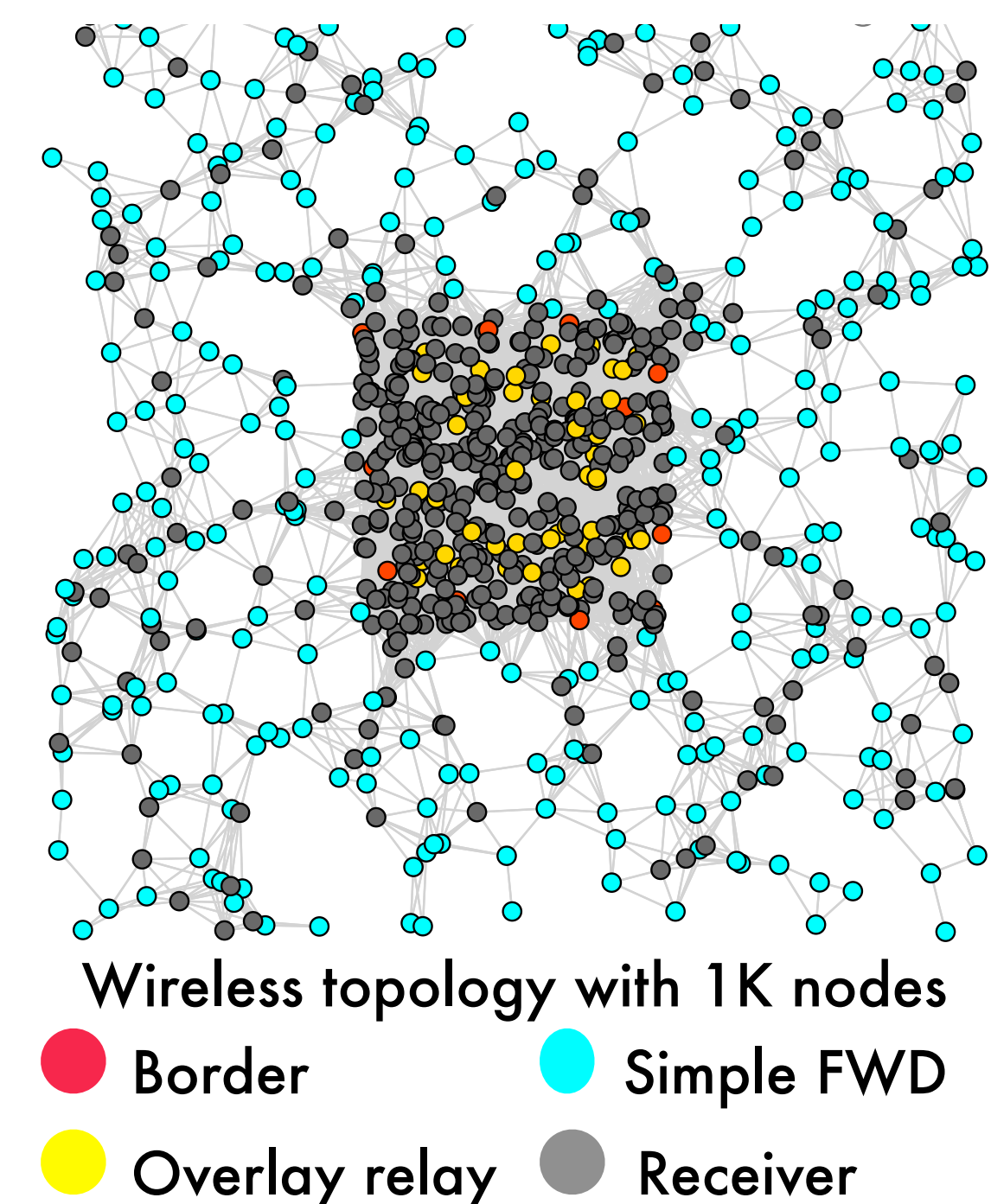
- hand-shake of nodes at border
- increase forwarding set of only one running protocol



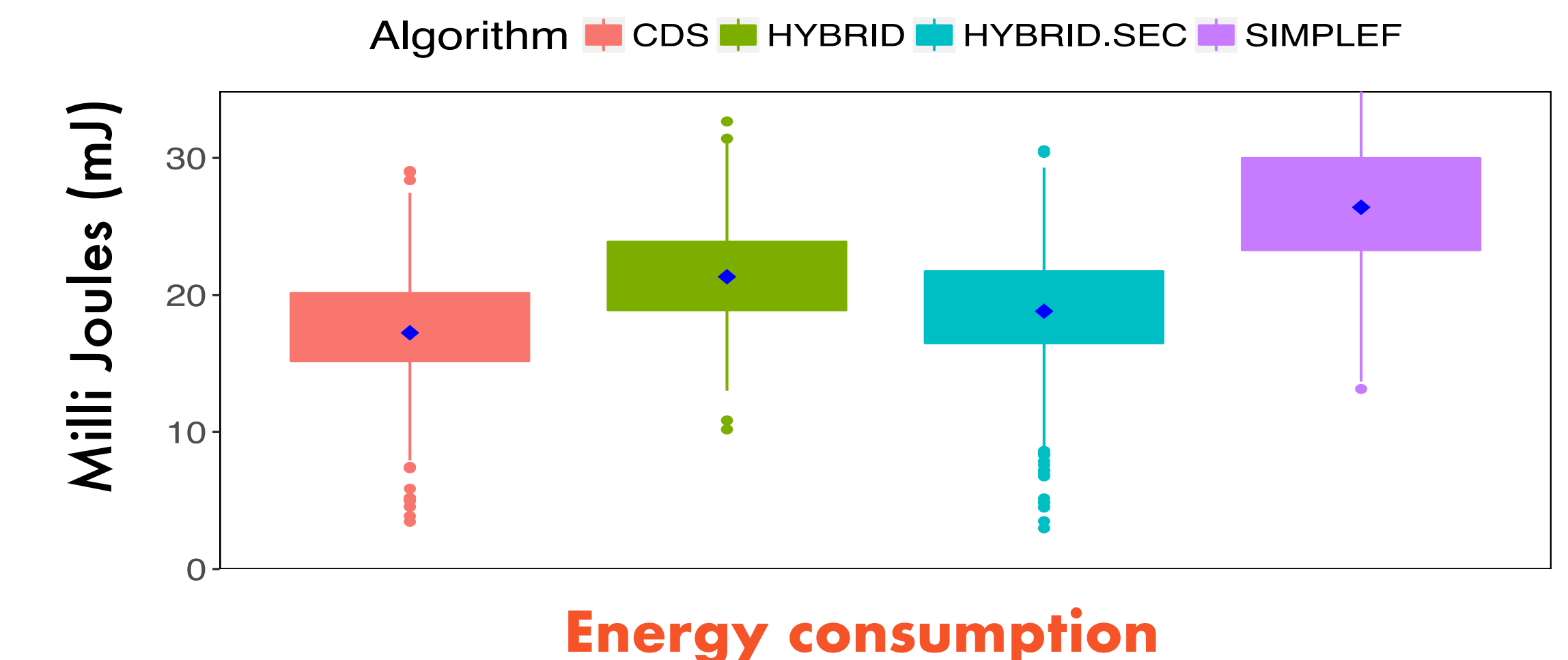
Experimental scenario

Simulations with Omnet++/INET

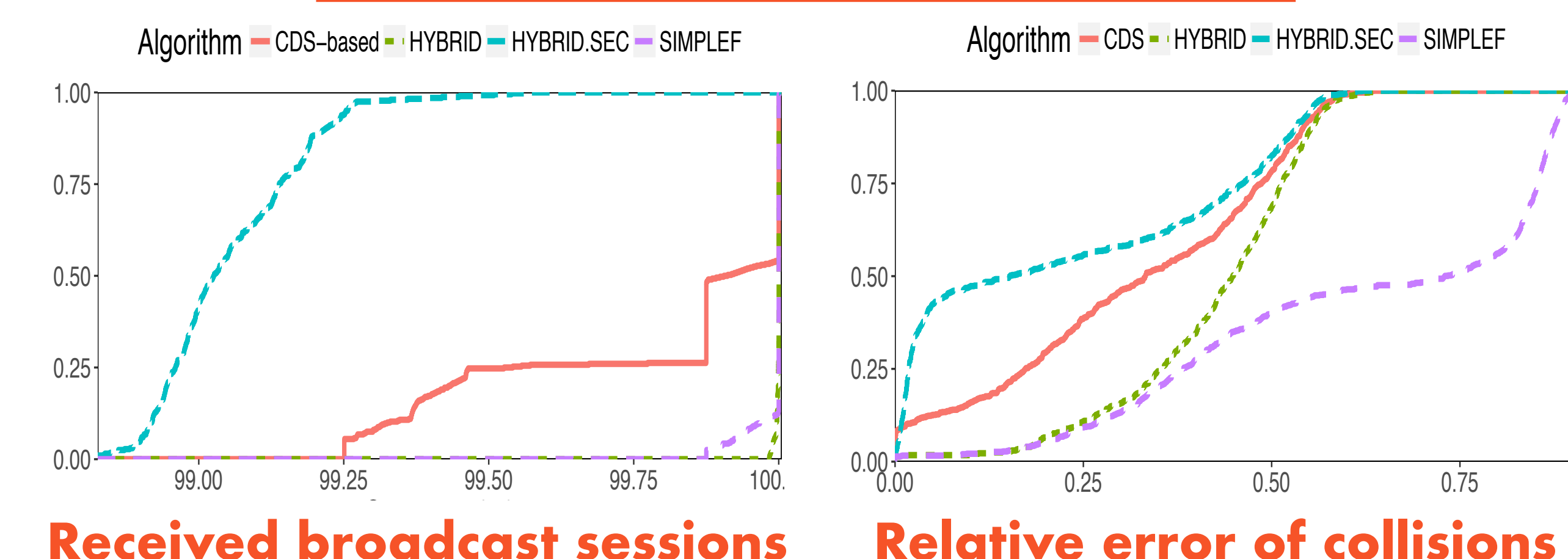
- transceiver commonly used in mobile phones
- devices with Wi-Fi (IEEE 802.11g) capabilities; CSMA/CA activated
- nodes follow the Levy-walk model to mimic how humans move
- heterogeneous regions: dense/dynamic & sparse/static
- source of broadcast chosen randomly over all nodes



Current Results



Cumulative Distribution Function over Nodes



References

- [1] S. Y. Ni, Y. C. Tseng, Y. S. Chen, and J. P. Sheu, "The broadcast storm problem in a mobile ad hoc network," in 5th Annual ACM/IEEE International Conference on Mobile Computing and Networking, MobiCom, 1999.
- [2] P. Ruiz and P. Bouvry, "Survey on Broadcast Algorithms for Mobile Ad Hoc Networks," ACM Comp. Sur., vol. 48, no. 1, pp. 1–35, 2015.
- [3] R. Carvajal-Gómez, I. Gonzalez-Herrera, Y. D. Bromberg, L. Réveillère, and E. Rivière, "Density and mobility-driven evaluation of broadcast algorithms for MANETs," in 2017 IEEE 37th International Conference on Distributed Computing Systems (ICDCS), June 2017, pp. 2308–2313.